

C# Entity Framework Cheatsheet

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1. Architektur

Core

- Entities, Enums, Value Objects
- Interfaces (Repositories, UnitOfWork)
- QueryResult/ReadModels

Persistence

- DbContext
- Entity Configurations (Fluent API)
- Repository Implementierungen
- UnitOfWork Implementierung

WebAPI

- DTOs
- Endpoint Mapping
- Validatoren
- Dependency Injection Registrierung

Import/Console

- Datei einlesen
- Daten transformieren
- Über Repositories speichern

2. Entity Template

```
public class EntityName : EntityObject
{
    public string RequiredText { get; set; } = string.Empty;
    public DateTime CreatedAt { get; set; }

    public int ParentId { get; set; }
    public ParentEntity Parent { get; set; } = null!;
}
```

3. Fluent API

Nutze Fluent API fuer DB-Regeln.

3.1. Configuration Template

```
public class EntityNameConfiguration : IEntityTypeConfiguration<EntityName>
{
    public void Configure(EntityTypeBuilder<EntityName> builder)
    {
        builder.HasKey(x => x.Id);

        builder.Property(x => x.RequiredText)
            .IsRequired()
            .HasMaxLength(100);

        builder.HasOne(x => x.Parent)
            .WithMany(p => p.Children)
            .HasForeignKey(x => x.ParentId)
            .OnDelete(DeleteBehavior.Restrict);

        builder.HasIndex(x => x.RequiredText);
    }
}
```

3.2. DbContext Template

```
public class ApplicationDbContext : DbContext
{
    public ApplicationDbContext(DbContextOptions<ApplicationDbContext> options) :
base(options) { }

    public DbSet<EntityName> EntityNames { get; set; } = null!;
    public DbSet<ParentEntity> ParentEntities { get; set; } = null!;

    protected override void OnModelCreating(ModelBuilder modelBuilder)
    {
        base.OnModelCreating(modelBuilder);
        modelBuilder.ApplyConfigurationsFromAssembly(typeof(ApplicationDbContext)
.Assembly);
    }
}
```

3.3. Migration Ablauf

Entweder über die Konsole oder über: Tools > Entity Framework Core > Add Migration / Update Database

```
dotnet ef migrations add InitialCreate --project .\Persistence\Persistence.csproj
--startup-project .\WebAPI\WebAPI.csproj

dotnet ef database update --project .\Persistence\Persistence.csproj --startup-project
.\WebAPI\WebAPI.csproj
```

Im Package Manager:

```
Add-Migration InitialCreate -Project Persistence -StartupProject Import

Update-Database -Project Persistence -StartupProject Import
```

Sqlite Installation:

```
dotnet add package Microsoft.EntityFrameworkCore.Sqlite
# Die Design-Tools für Migrationen (Add-Migration, etc.)
5 dotnet add package Microsoft.EntityFrameworkCore.Design
```

4. Repository + UnitOfWork Muster

4.1. Contracts in Core

```
public interface IEntityRepository : IGenericRepository<EntityName>
{
    Task<IList<EntityOverview>> GetOverviewAsync(string? filter, bool? onlyActive);
}

public interface IUnitOfWork : IBaseUnitOfWork
{
    IEntityRepository Entities { get; }
    IAnotherRepository Others { get; }
}
```

4.2. UnitOfWork in Persistence

```
public class UnitOfWork : BaseUnitOfWork, IUnitOfWork
{
    public UnitOfWork(ApplicationDbContext dbContext) : base(dbContext)
    {
        Entities = new EntityRepository(dbContext);
        Others = new AnotherRepository(dbContext);
    }

    public IEntityRepository Entities { get; }
    public IAnotherRepository Others { get; }
}
```

4.3. Query-Regeln für gute Read-Methoden

- Für reine Leseabfragen: `AsNoTracking()`
- Erst filtern, dann projizieren
- In Repository nur Datenlogik, keine API-Logik
- Rückgabe für Listen: `IList<T>` oder `List<T>` konsistent halten

5. ReadModels / Overview-Objekte

Nutze eigene Typen für Auswertungen statt Entities 1:1 nach außen zu geben.

```
public record EntityOverview(  
    int Id,  
    string DisplayName,  
    bool IsActive,  
    int RelatedCount  
);
```

5.1. Typisches Query-Template

```
public async Task<IList<EntityOverview>> GetOverviewAsync(string? filter, bool?  
onlyActive)  
{  
    var query = _dbContext.Entities.AsNoTracking().AsQueryable();  
  
    if (!string.IsNullOrEmpty(filter))  
        query = query.Where(x => x.Name.Contains(filter));  
  
    if (onlyActive == true)  
        query = query.Where(x => x.ActiveUntil == null || x.ActiveUntil >= DateTime  
.Today);  
  
    return await query  
        .Select(x => new EntityOverview(  
            x.Id,  
            x.Name,  
            x.ActiveUntil == null || x.ActiveUntil >= DateTime.Today,  
            x.Children.Count))  
        .OrderBy(x => x.DisplayName)  
        .ToListAsync();  
}
```


6. API Design Muster

6.1. DTO statt Entity direkt verwenden

```
public record EntityDto(int Id, string Name, string Type);
```

6.2. Mapper-Funktionen (DTO $\leftrightarrow$ Entity)

Halte Mapping zentral in der Endpoint-Klasse (oder in einer separaten Mapper-Klasse), damit Create/Update/Get konsistent bleiben.

```
private static EntityName ToEntity(EntityDto dto)
{
    return new EntityName
    {
        Id = dto.Id,
        Name = dto.Name,
        Type = dto.Type
    };
}

private static EntityDto? ToDto(EntityName? entity)
{
    if (entity is null)
        return null;

    return new EntityDto(entity.Id, entity.Name, entity.Type);
}

private static IList<EntityDto>? ToDto(IList<EntityName>? entities)
{
    if (entities is null)
        return null;

    return entities.Select(x => ToDto(x!)).ToList();
}
```

Mapper-Regeln:

- `ToDto(Entity?)` gibt `null` zurueck, wenn Entity nicht gefunden wurde
- Listen immer ueber `Select(...).ToList()` abbilden
- Im `POST` und `PUT` keine Roh-Entities aus dem Request verwenden

6.3. Endpoint-Gruppe aufbauen

```
public static void MapEntityEndpoints(this IEndpointRouteBuilder app, string
baseRoute)
{
    var route = app.MapGroup(baseRoute).WithTags("Entity");

    // route.MapGet...
    // route.MapPost...
}
```

6.4. Request-Parameter in Minimal APIs

- Path-Parameter: `"/{id:int}"` + Lambda-Argument `int id`
- Query-Parameter: `[FromQuery(Name = "filter")] string? filter`
- Body-Parameter: komplexe Typen wie `EntityDto dto`
- Services per DI: z.B. `IUnitOfWork uow`

```
route.MapGet("/overview", async (
    [FromQuery(Name = "filter")] string? filter,
    [FromQuery(Name = "onlyActive")] bool? onlyActive,
    IUnitOfWork uow) =>
{
    var data = await uow.Entities.GetOverviewAsync(filter, onlyActive);
    return Results.Ok(data);
});
```

6.5. GET alle (GET /entities)

Zweck: alle Datensätze laden (read-only).

```
route.MapGet("", async (IUnitOfWork uow) =>
{
    var dtos = ToDto(await uow.Entities.GetNoTrackingAsync());
    return Results.Ok(dtos);
})
.WithName("GetEntities")
.Produces<List<EntityDto>>(StatusCodes.Status200OK);
```

6.6. GET by id (GET /entities/{id})

Zweck: genau einen Datensatz per Path-Parameter laden.

```

route.MapGet("/{id:int}", async (int id, IUnitOfWork uow) =>
{
    var dto = ToDto(await uow.Entities.GetByIdAsync(id));

    if (dto is null)
    {
        return Results.Problem(
            statusCode: StatusCodes.Status404NotFound,
            title: "Entity not found",
            detail: $"No entity found with ID {id}");
    }

    return Results.Ok(dto);
})
.WithName("GetEntity")
.Produces<EntityDto>(StatusCodes.Status200OK)
.ProducesProblem(StatusCodes.Status404NotFound);

```

6.7. POST create (POST /entities)

Zweck: neuen Datensatz anlegen.

```

route.MapPost("", async (EntityDto dto, IUnitOfWork uow) =>
{
    if (dto.Id != 0)
    {
        return Results.Problem(
            statusCode: StatusCodes.Status400BadRequest,
            title: "Invalid request",
            detail: "The ID in the request body must be 0");
    }

    var entity = ToEntity(dto);
    await uow.Entities.AddAsync(entity);
    await uow.SaveChangesAsync();

    var createdDto = ToDto(await uow.Entities.GetByIdAsync(entity.Id));
    return Results.Created($"{baseRoute}/{entity.Id}", createdDto);
})
.WithValidation<EntityDto>()
.WithName("AddEntity")
.Produces<EntityDto>(StatusCodes.Status201Created)
.ProducesProblem(StatusCodes.Status400BadRequest);

```

6.8. PUT update (PUT /entities/{id})

Zweck: bestehenden Datensatz ersetzen/aktualisieren.

```

route.MapPut("/{id:int}", async (int id, EntityDto dto, IUnitOfWork uow) =>
{
    if (id != dto.Id)
    {
        return Results.Problem(
            statusCode: StatusCodes.Status400BadRequest,
            title: "Invalid request",
            detail: "The ID in the URL does not match the ID in the request
body");
    }

    var entity = await uow.Entities.GetByIdAsync(id);

    if (entity is null)
    {
        return Results.Problem(
            statusCode: StatusCodes.Status404NotFound,
            title: "Entity not found",
            detail: $"No entity found with ID {id}");
    }

    entity.Name = dto.Name;
    entity.Type = dto.Type;

    await uow.SaveChangesAsync();
    return Results.NoContent();
})
.WithValidation<EntityDto>()
.WithName("UpdateEntity")
.Produces(StatusCodes.Status204NoContent)
.ProducesProblem(StatusCodes.Status400BadRequest)
.ProducesProblem(StatusCodes.Status404NotFound);

```

6.9. DELETE (DELETE /entities/{id})

Zweck: Datensatz löschen.

```

route.MapDelete("/{id:int}", async (int id, IUnitOfWork uow) =>
{
    var entity = await uow.Entities.GetByIdAsync(id);

    if (entity is null)
    {
        return Results.Problem(
            statusCode: StatusCodes.Status404NotFound,
            title: "Entity not found",
            detail: $"No entity found with ID {id}");
    }
}

```

```
uow.Entities.Remove(entity);  
await uow.SaveChangesAsync();  
  
return Results.NoContent();  
})  
.WithName("DeleteEntity")  
.Produces(StatusCodes.Status204NoContent)  
.ProducesProblem(StatusCodes.Status404NotFound);
```

7. Validierung (FluentValidation)

Halte Regeln im API-Projekt, nicht in Endpoints verstreut.

```
public class EntityDtoValidator : AbstractValidator<EntityDto>
{
    public EntityDtoValidator()
    {
        RuleFor(x => x.Name)
            .NotEmpty()
            .MinimumLength(3)
            .MaximumLength(100)
            .Matches("^[a-zA-Z0-9 ]+$")
            .WithMessage("Name can only contain letters, numbers and spaces.");

        RuleFor(x => x.Type)
            .NotEmpty();
    }
}
```

8. WebAPI Program Setup

```
var builder = WebApplication.CreateBuilder(args);

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();
builder.Services.AddProblemDetails();

builder.Services.AddCors(options =>
{
    options.AddDefaultPolicy(policy =>
    {
        policy.AllowAnyHeader().AllowAnyOrigin().AllowAnyMethod();
    });
});

builder.Services.AddScoped<IUnitOfWork, UnitOfWork>();
// Falls UnitOfWork weitere Services benötigt, diese hier ebenfalls registrieren (z.B.
// IAnotherRepository)

var configuration = ConfigurationHelper.GetConfiguration();
var connectionString = configuration.GetConnectionString("DefaultConnection")
    ?? throw new InvalidOperationException("Connection string 'DefaultConnection' not
found.");

builder.Services.AddDbContext<ApplicationDbContext>(options =>
    options.UseSqlServer(connectionString));

builder.Services.AddValidatorsFromAssemblyContaining<Program>();

var app = builder.Build();

app.UseExceptionHandler();
app.UseStatusCodePages();
app.UseCors();

if (app.Environment.IsDevelopment())
{
    app.UseSwagger();
    app.UseSwaggerUI();
}

app.UseHttpsRedirection();

// Minimal API Endpoints (Falls gewünscht, ansonsten in Extension-Methoden kapseln)
// einfach vom Template oben nehmen und statt route direkt app verwenden
app.MapGet("/ping", () => "pong")
    .WithName("Ping")
    .WithTags("Health");
```

```
// Endpoint-Mapping in Extension-Methoden kapseln, damit Program.cs sauber bleibt,  
falls mehrere Entities/Endpunkt-Gruppen existieren  
app.MapEntityEndpoints("/api/entity");  
app.MapAnotherEntityEndpoints("/api/another-entity");  
  
app.Run();
```


9. CSV Import

```
var builder = Host.CreateApplicationBuilder(args);

var configuration = builder.Configuration;
var connectionString = configuration.GetConnectionString("DefaultConnection")
    ?? throw new InvalidOperationException("Connection string 'DefaultConnection' not found.");

builder.Services
    .AddDbContext<ApplicationDbContext>(options =>
        options.UseSqlServer(connectionString))
    .AddScoped<IUnitOfWork, UnitOfWork>();

var host = builder.Build();

Console.WriteLine("Migrate Database");

using (var scope = host.Services.CreateScope())
{
    var uow = scope.ServiceProvider.GetRequiredService<IUnitOfWork>();
    await uow.DeleteDatabaseAsync();
    await uow.MigrateDatabaseAsync();
}

Console.WriteLine("Import Data");

using (var scope = host.Services.CreateScope())
{
    var uow = scope.ServiceProvider.GetRequiredService<IUnitOfWork>();

    var csvData = await new CsvImport<TModel>().ReadAsync("ImportData/Source.csv");
    var cache = (await uow.Entities.GetAsync()).ToDictionary(e => e.UniqueKey, e => e);

    // LINQ-Pipeline statt foreach
    var pending = csvData.Select(async item =>
    {
        if (!cache.TryGetValue(item.Identifier, out var entity))
        {
            entity = new TEntity { PropertyA = item.ValueA, PropertyB = item.ValueB };
            await uow.Entities.AddAsync(entity);
            cache.Add(item.Identifier, entity);
        }
        entity.PropertyA = item.ValueA;
        return entity;
    }).ToList(); // .ToList() erzwingt die Iteration durch die Daten

    await Task.WhenAll(pending);
}
```

```
}
```

Falls das nicht funktioniert, hier noch eine weitere Variante:

```
using (var scope = host.Services.CreateScope())
{
    var uow = scope.ServiceProvider.GetRequiredService<IUnitOfWork>();

    var entitiesCsv = await new CsvImport<Entity>().ReadAsync(
        "ImportData/Entity.csv");

    var entities = entitiesCsv.Select(x =>
    {
        return new Entity
        {
            PropertyA = x.PropertyA,
            PropertyB = x.PropertyB,
        };
    }).ToList();

    await uow.Entities.AddRangeAsync(entities);
    await uow.SaveChangesAsync();
}
```

10. Unit Tests

10.1. XUnit

```
using Xunit;
using YourProject.Services;
namespace YourProject.Tests;

public class YourServiceTests
{
    private readonly IYourService _yourService;

    public YourServiceTests()
    {
        _yourService = new YourService();
    }

    [Fact]
    public void YourMethod_ShouldReturnExpectedResult()
    {
        // Arrange
        var input = ...;

        // Act
        var result = _yourService.YourMethod(input);

        // Assert
        Assert.Equal(expectedResult, result);
    }
}
```

10.2. NUnit

```
using NUnit.Framework;
using YourProject.Services;
namespace YourProject.Tests;

public class YourServiceTests
{
    private IYourService _yourService;

    [SetUp]
    public void Setup()
    {
        _yourService = new YourService();
    }
}
```

```

[Test]
public void YourMethod_ShouldReturnExpectedResult()
{
    // Arrange
    var input = ...;

    // Act
    var result = _yourService.YourMethod(input);

    // Assert
    Assert.AreEqual(expectedResult, result);
}
}

```

10.3. FluentAssertions

```

using FluentAssertions;
using Xunit;
using YourProject.Services;
namespace YourProject.Tests;

public class YourServiceTests
{
    private readonly IYourService _yourService;

    public YourServiceTests()
    {
        _yourService = new YourService();
    }

    [Fact]
    public void YourMethod_ShouldReturnExpectedResult()
    {
        // Arrange
        var input = ...;

        // Act
        var result = _yourService.YourMethod(input);

        // Assert
        result.Should().Be(expectedResult);
    }
}

```